Ref: Elcon/M/s. R.B.Construction/DD/2425-149/05 **Date**: 28.11.2025

To,
Executive Engineer,
Drainage Department,
KDMC, Kalyan (West).

Sub: Design of drainage disposal system for Proposed Resi. & Comm. on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje- Chikanghar, Tal- Kalyan, Dist.-Thane.

Dear Sir,

We are submitting design of drainage disposal system for Proposed Resi. & Comm. on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane. The drainage system has been designed for the BLDG.-1 population of 1042 person STP capacity proposed is 125 kld KDMC benchmark levels has been followed for the same.

Thanking You,
For Elcon Engineers

SUBODH GOTHOSKAR

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ELCON ENGINEERS DOMBIVLI

Authorised Signatory

PROPOSED RESI. & COMM. BUILDING ON PLOT BEARING S. No. 105/8/2, 97/1/9, 98/2/5 OF MOUJE - CHIKANGHAR, TAL. - KALYAN, DIST. - THANE.

NAME OF DOCUMENT		KDMC DBR	
NO. OF DOCUMENT		1	
DATE		01.12.2025	
CLIENT	STRUCTURAL CONSULTANT	MEP CONSULTANT	ARCHITECT
M/S. R.B. CONSTRUCTION		ELCON ENGINEERS	SAGAR ASSOCIATES

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Report Prepared By : Mr. Subodh Gothoskar

ELCON ENGINEERS DOMBIVLI

SUBODH GOTHOSKAR

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1.1 CODES AND STANDARDS:

1. Following codes, standards and reference are being used for designing the Plumbing System for the (M/s. R. B. Construction)

a) National Building Code of India, 2016.
b) Handbook on Water Supply and Drainage.
c) Plumbing Engineering Services Designer Code of the British Institute of Plumbing.
d) Uniform Illustrated Plumbing Code - India.
e) Municipal Code of Practice.
f) American Society of Plumbing Engineers.
g) American Society of Mechanical Engineers.

1.2 GENERAL NOTES:

1. The project includes BLDG.-1 (GR.+Mezzanine+1ST & 2ND Amenity Fl.+ 3RD To 28TH Resi. FL.) ,
2. For water supply system for flats, we propose conventional gravity system.
3. Plumbing shafts shall be provided adjacent to each Toilet and kitchen.
    a. All floors Toilet slabs shall be sunken.

1.3 WATER SOURCE & IT'S REQUIRMENT:

Water supply requirements have been estimated as per Residential Bldg. based on the number users as per UDCPR with 5 person per tenant with adhered to the guidelines in the National Building Code of India, 2016 as per Table 1 (Clause 4.1.2)

1. Source of water is from Bore well/ Tanker & as per the water quality further water Treatment will be done to use this water for domestic application.
2. Provision is made for connecting municipal water supply line.
3. Total per day peak water requirements for the project for dry season shall be as belo

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WATER DEMAND STATEMENT FOR (PROJECT DETAILS)											
Sr. No.	Bldg. Name	Unit	No. of. FLR LVL	No. of Bldg.	No. of Per Son/ Unit	Total Nos. of Person	Average daily water REQD / Person 135 LPCD	U.G. Water Tank		O.H. Water Tank 0.5 day	
								Domestic 90 LPCD	Flushing 45 LPCD	Domestic	Flushing
A	**Bldg.-1**										
1.1	Total No. of Floor (GR.+Mezzanine+1ST & 2ND Amenity FL.+ 3RD To 28TH Resi. FL.)										
a	Flats	151	26	1	6	906	135	81540	40770	40770	20385
b	Shops + Mezz.	846.63 sq.mt	01		10.00sq.mt/ person	85	45	2125	1700	1063	850
c	Offices	506.95 sq.mt.	01		10.00sq.mt/ person	51	45	1275	1020	638	510
	Total of Daily Water Demand for Flats (1 Day Capacity in Ltrs)						128430	84940	43490	42471	21745
	U.G. Tank (1.5 Days Capacity in Cum) Ltrs / Day							128	66	64	33

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C	Total No. of Unit	151 Flat + 09 Shops + 08 Offices
D	Total Population Residential	906 No. of Persons
	Total Population Commercial	136 No. of Persons
E	Total of Daily Water Required	128430.00 Ltrs.

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1. Total fresh domestic water demand = 85 cum / day.
2. Total flushing water demand = 44 cum / day.
3. The flushing water & landscape water requirement shall be fulfilled by the treated water produced from the common STP of 125 KLD in the project.
4. Instead of grey water treatment plant / ETP, sewer water treatment plant / STP is proposed, as it can treat the black water from the water closet & Urinals along with the grey water & is better than Grey water plant / ETP.

TOTAL SEWER QUANTITY TO COMMON STP FOR (OF TOTAL 90% DOMESTIC WATER & TOTAL FLUSHING WATER)	121
SAY STP IN KLD	125
NOTE -1 : THIS STP WATER TANK WILL CATER REQUIRED CAPACTY	
NOTE - 2 : THIS STP WATER TANK CAPACITY AS PER 135 LPCD (90 LTRS DOM. + 45 LTRS	

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1.4 WATER STORAGE :-

1. The Domestic water requirement will be supplied through the bore well / Tanker. The flushing & landscape water requirement will be full filled from the STP treated water.
2. Summary of the water storage tank proposed is as follow:-

	SUMMARY	U.G. TANK WITH 1.5 DAYS STORAGE IN CUM.				O H TANK WITH 0.5 DAYS STORAGE IN CUM	
	U G TANK & O H TANK IN CUM / DAY	DOMESTIC	DIMENSION (L X B X H)	FLUSHING	DIMENSIN (L X B X H)	DOMESTIC	FLUSHING
1.0	Fire U. G. Tank considering common for BLDG.- 1 Height as per NBC 2016	WING-A = 150 (8.35 x 6.0 x 3.0)					
2.0	Residential (BLDG.-1)	128	7.15x6.0x3.0	66	3.70x6.0x3.0	64	33
2.1	Fire tank OHT	----			25		

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1.5 WATER DISTRIBUTION

1.5.1 Water Distribution System:

1 The incoming main lines from Bore well lines shall be fed into underground fire reserve tank from where it shall be allowed to overflow in domestic storage tank, also connection made for domestic water tanks of other buildings without the UG fire tank on that building. Recycled water from STP shall be used for flushing/landscape purpose which will be pumped to the flushing water tanks of each building.

2 All domestic water tanks will be having set of submersible pumps which shall transfer the water from domestic UGT to domestic OHT of each building. Similarly all flushing water tanks will be having set of submersible pumps which shall transfer the water from flushing UGT to flushing OHT of each building.

3 From the overhead tanks, distribution of domestic water shall be done by adopting the gravity system by providing down take pipes from closed grid loop at terrace to all user points/ fixtures. The pipe size shall be based on fixture unit calculations as per NBC 2016 standard.

4 Water distribution for water supply shall be designed on principle of zoning to ensure minimum residual pressure of 1.2 to 2.1 Bar at all user outlets.

5 In order to get adequate flow for top two floors, separate looping shall be done from taking another outlet from OH tank.

6 Pressure reducing valves shall be provided for the lower floors in order to adequate pressure as per NBC-2016.

7 For Flushing of WCs, a separate water supply system shall be provided, consisting of separate Overhead tanks on Terrace, and gravity supply through separate down-takes. This separate system for flushing shall ensure that the treated effluent water from the STP is re-used. Emergency connection for normal potable Water supply shall also be made to this system.

8 The distribution network shall be provided with adequate number of control valve/ NRV for ensuring continuous uninterrupted supply of water to individual user's outlet. Ball/Butterfly valves of approved make shall be provided as close to the underground tank on every outlet pipe from the storage tank, except the overflow pipe. Overflow and vent pipes shall terminate with mosquito proof grating.

9 The overflow pipe shall be so placed as to allow the discharge of water being readily seen. A stop valve shall also be provided in the inlet water connection to the tank.

10 All inlets, outlets, washouts, vents, ball cocks, overflows control valves and all such other accessories, piping connections including level indicator shall be provided for underground water storage tanks.

11 The pipe sizing shall be based on fixture unit calculation as per NBC 2016 standard.

12 Color coding for various water supply pipes shall be ensured for clear identification.

1.5.2 The design of water supply system following Fixture units (FU)

Will be considered as per NBC-2016 Table 2 (Clause 4.7.3.1): -

a) Water closet with Flush valve	-	6 FU (each)
b) Wash basin (domestic)	-	1 FU (each)
c) Urinal with Flush value	-	3 FU (each)
d) Ablution tap	-	1 FU (each)
e) Shower	-	2 FU (each)
f) Kitchen sink (domestic)	-	2 FU (each)

1.5.3 Pipe connections (dia.) to various fixtures will be as NBC-2016 Table 3 (Clause 4.7.3.1) follows.

a) Water closet with flush valve	-	25/32 mm
b) Wash basin	-	15 mm

c) Urinal	-	15/20 mm
d) Bib tap	-	15 mm
e) Faucet	-	15 mm
f) Kitchen sink	-	15 mm/20mm

1.6 SANITARY FIXTURES AND FITTINGS (As approved by Architect)

1. All sanitary wares shall be white vitreous China and of standard quality and make.
2. In Building, European pattern wall hung EWC with flush tank has been considered in all toilets.
3. The low flow bib taps should be considered.
4. Kitchen Sinks shall be of stainless steel with drain board along with CP waste coupling and bottle trap shall be provided. The low flow sink cock should be considered.

1.7 SANITARY DRAINAGS (SOIL AND VENT):

1. All sanitary fixtures will be connected to floor trap (or) 'P' trap with Effective water seal traps.
2. The installation will be designed as a fully ventilated two-pipe system with separate soil and waste stacks. Soil stacks will be cross-vented to a separate vent/anti-syphon age stack.
3. Waste appliances will be connected to deep-seal floor drains or "P' trap and no separate venting of appliances is proposed.
4. Deep seal traps with access caps will be provided at base of waste stacks.
5. The location of stacks shall be coordinated with other services.
6. The sewerage and drainage system shall be designed keeping in view of natural gradient at site, layout of different facilities in block, level of road and other services in vicinity etc.
7. The branch and Down stack pipe sizes will be designed on the basis of FU calculations based on the number of fixture units as per N.B.C of India 2016 Table no 16 (Clause 4.5.2.4.2.1)
8. Total drainage load to connect to sewer line will be calculated based on the number of fixture units specified in N.B.C, 2016 Table no 20 (Clause 4.5.3.5.1).

1.8 RAIN WATER HARVESTING RECHARGE PIT

1. Roof drain pipes will be designed considering peak intensity of rainfall of 100mm/hr.
2. Rain water down take pipes shall be installed in plumbing shafts and / or external to the building.
3. External down pipes shall be suitably located based on architectural provisions.
4. Roof outlets will be vertical with bell mouth and khurra with dome gratings. Area of openings onthe grating shall be 150% of area of down take pipe.
5. Rain water will be collected by down take pipes into the recharge pit. The rain water harvesting-Recharge pits is proposed to increase the ground water which shall increase the bore well water level to supply the water requirement for domestic purpose during & after monsoon.

RUNOFF CALCULATION							
A	**Pre development Runoff**						
Sr. no.	**Area**	**Surface type**	**Runoff coefficient**	**Storm Catchment (Area in sq. m)**	**Impervious Area in sq. m**	**15 min peak Intensity of rainfall mm/day**	**Discharge (Q)cum/day**
1	Land	Barren Land	0.3	3676.00	1103	42	46
		Total area		3676.00	1103		46
					Total discharge cum/day		**46**
B	**Post development Runoff**						
Sr. no.	**Area**	**Surface type**	**Runoff coefficient**	**Storm Catchment (Area in sq. m)**	**Impervious Area in sq. m**	**15 min peak Intensity of rainfall mm/day**	**Discharge (Q)cum/day**
1	BLDG.-1	Roof	0.9	581.75	524	42	22
2	RG Area	Unpaved RG	0.3	0.00	0.00	42	00
3	Hard Scape	Roads & Hard Paved	0.75	3094.25	2321	42	97
		Total area		3676.00	2844		119
					Total discharge cum/day		**119**
	Difference of pre development runoff and post development run off will be					**73**	cum/day
	22.00 Cu. M water will be collected from roof & same will be given to borewell for recharge.						

6. The overflow from the recharge pits shall be connected to the storm water recharge trenches.
7. The average annual rainfall intensity considers for rain water harvesting pits is considering 0.1 m/hr

1.9 STORM WATER DRAINAGE :

1. Storm water drain (SWD) channel is considered to drain off the rain water from the complete plot.
2. This internal plot SWD channel is connected with No. of storm water recharge trenches.
3. The storm water calculation is as below

STORM WATER DRAIN SIZE CALCULATIONS			
Discharge - Q = R*A*P M^3/hr where,			
R = Maximum Intensity of Rainfall as 0.1 M/hr			
A = Area in M^2			
P = Impermeability factor			
NO.	**DESCRIPTION**	**UNIT**	**QTY.**
1	**DISCHARGE OF UNPAVED AREA (RG AREA)**		
a.	Maximum Intensity of Rainfall (R)	M/hr	0.10
b.	Area (A)	M^2	0.00
c.	Impermeability factor (P)	-	0.50
	Discharge (Q1 = R*A*P)	M^3/hr	0.00
		M^3/sec	0.01
2	**DISCHARGE OF PAVED AREA**		
a.	Maximum Intensity of Rainfall (R)	M/hr	0.10
b.	Area (A)	M^2	4654.00
c.	Impermeability factor (P)	-	0.95
	Discharge (Q2 = R*A*P)	M^3/hr	442.13
		M^3/sec	0.12
3	**DRAIN SIZE CALCULATIONS FOR DRAINS**		
a.	Total discharge from RG area (Q1)	M^3/sec	0.00
b.	Total discharge from Paved area (Q2)	M^3/sec	0.12
	TOTAL ACTUAL DISCHARGE (Q)	**M^3/sec**	**0.12**
4	**DISCHARGE DESIGN**		
a.	Assume drain channel width (W)	M	0.60
b.	Assume drain channel depth (D)	M	0.40
c.	Area of drain channel (A = W*D)	M^2	0.24

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d.	Wetted Perimeter (P = W + 2D)	M	1.40
e.	Wetted Radius (R = A / P)	M	0.17
f.	Slope 1:500 (S)	-	0.002
g.	Frictional co-efficient (n)	-	0.013
h.	Discharge by Manning's Formula [V = (1/n) x (R)2/3 x (S1/2)]	-	1.062
	TOTAL DESIGN DISCHARGE (Q = A x V)	**M³/sec**	**0.25**
	PROPOSED DRAIN CHANNEL WIDTH (W)	M	0.60
	PROPOSED DRAIN CHANNEL DEPTH (D)	M	0.40
Since Design Discharge is greater than Actual Discharge, the size of the drain designed is okay.			
Catchment area as adjacent plot in either direction should be added in this Calculation if it is falling in the said plot.			

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1.10 SEWAGE WATER TREATMENT PLAN

GENERAL DESCRIPTION BASED ON MBBR TECHNOLOGY

i. **Preliminary Treatment — Screening and Oil & Grease Removal**

Sewage generated is collected in common Drain. Common Drain Pipe is further connected to Inlet Chamber of STP. Sewage is further passed thru 10 mm manual Screen for removal of suspended particles and further it passes through Corrugated Plate Interceptor for Oil and Grease removal. The 'Floating Free Oil' shall be removed manually from oil and grease Chamber. The Screenings and Floating oil shall be removed periodically. Keeping in view the Flow variation at the Process Plant, Equalization Tank is designed at Downstream of Oil and Grease Chamber. Equalized sewage is kept in suspension by using air blower and it is pumped to MBBR Reactor 1&2.

ii. **Secondary Treatment- Moving Bed Bio-Reactor (MBBR) System & Secondary Settling Tanh (SST)**

Sewage Pumped from Equalization Tank flows to MBBR Reactor. The proposed biological process is the MBBR system, which utilizes attached growth grown on biomass carriers or media. Aeration will be achieved by coarse bubble diffusers, installed at the reactor bottom in the required pattern. Unique airlift hydraulics will assure mixing of biomass carriers. A media retention screen will be installed at outlet of each stage in order to maintain the carriers inside the reactor, whereas mixed liquor may flow unobstructed. The media retention screens are continuously cleaned by the mixing flow of water, air bubbles and biomass carriers. The treated wastewater will flow from the biological reactor to a SST for solid-liquid separation. Excess Sludge generated
from SST shall be withdrawn to Sludge holding Sump.

iii. **Tertiary Treatment — Disinfection and Filtration**

The Treated Water from SST flows to Clarified Water Tank, where Hypochlorite is dosed for disinfection purpose. Further treated sewage is pumped to PSF and ACF Filters. It will have sand and carbon as Filter would be provided complete with filter backwash arrangement to ensure reduction of Suspended solids & turbidity at the outlet of the PSF and ACF Filters. When the pressure Drop across the filter increases a certain limit, Filter shall be run under backwash mode each filter at a time. Filter Backwash shall be recycled back at the equalization Tank. Filtered water is to be discharged in Existing Drain.

iv. **Sludge Dewatering System - Centrifuge with Centrifuge Feed Pump.**

Sludge from SST flows to Sludge Collection Tank by Gravity. Sludge is fed to centrifuge by means of Centrifuge Feed Pump.

Sewage Characteristics	**Inlet**	**Outlet**
BOD (ppm)	250 -300	10
COD (ppm)	400-500	30
TSS (ppm)	150	< 5
Oil and Grease	<2O	< 10
Temperature (in degree Celsius)	25 — 30	25-30
Online PH Correction system	5.5-9	6.5-8

1.10.2 PROCESS DETAILS

Bar Screen :

Raw sewage from the source is usually received into the bar screen chamber by gravity. Screen provided will remove all floating and big size matter such as plastic bottles, polythene bags, glasses, stones, etc., which may otherwise choke the pipeline and pumps.

Oil and Grease Trap (Civil Construction)

If the sewage generated includes maximum quantity from kitchen and canteen, there is a possibility of higher concentrations of oil and grease in the raw sewage. It needs to be removed before biological treatment as it otherwise may cause problems for biological treatment. Usually, a small civil construction tank with a baffle wall and slotted oil pipe skimmer is provided. The oil and grease removed by gravity floats to the surface, which is removed by the oil skimmer (by client).

Equalization Tank (Civil Construction)

Usually, sewage generation is more during morning hours and evening hours. Visually no sewage is generated during night hours. Any biological system needs constant feed for bacteria to work efficiently. Hence, it is important to put an equalization tank to collect the excess flow during peak hours and feed sewage in lean hours. A typical equalization tank has a capacity of 8 - 12 hours of average flow rate. The tank is generally of civil construction by client. Provision of air grid is to be made for thoroughly mixing the sewage to make it of homogenous quality and to keep the suspended matter in suspension and to avoid septic conditions.

Transfer of Sewage

IEI's scope starts from transfer of sewage from Equalization Tank to MBBR tank. The distance of Transfer should not exceed beyond 5 meter. The transfer pump can be either submersible or non-submersible type for this application. However IEI has considered centrifugal non-submersible type.

The layout shall be as per IEI standard.

The pump should not run dry and Client to ensure sufficient sewage is available in the Equalization Tank.

Moving Bed Bio Reactor (MBBR)

Moving Bed Bio Reactor (MBBR) as the name indicates consists of floating media of various Shapes and sizes. The main objective of adding this media is to make available more surfaces Area for bacteria to grow on, thereby maintaining and retaining maximum possible bacterial Population in a limited volume. The MBBR media material allows biomass concentration of 20 — 40 Kg.s/m.3 material. Thus, FMR consists of combination of biomass in attached as well as suspended form. High concentration of biomass enables reduction of aeration tank and in turn reduction in overall cost. Volume of the media shall vary from 6 to 25 % based on the Concentration of organic matter. Another main feature of the MBBR is its compactness. The MBBR consists of biological system for removal of organic matter (BOD, COD), lamella for clarification and chlorine contact tank for disinfection. As all units are placed inside a single tank, it saves space and also increases operational ease.

In MBBR, raw sewage enters at the top of the tank. Air is introduced at the bottom of the tank

through fine bubble diffusers. Media will be in suspension because of the turbulence created by the air. The bacteria required for the oxidation of the organic matter is attached to the media and some part is suspended in the tank. After oxidation, the bacteria grow in number and need to be separated from the aeration tank liquor. The lamella section inside the MBBR helps in clarification and separation of the bacteria (sludge) and clear overflow flows into chlorine contact tank. Lamella plates helps in increasing the settling area and removing the particles effectively in a smaller plan area. In chlorine contact tank, Sodium hypo Chlorite (Na0C1) is added for disinfecting the clarified sewage. Baffle plates are provided to make better contact. The chlorinated treated sewage then flows out of MBBR either for further treatment or for disposal.

Treated water collection Tank :

The treated water collection tank can be of civil construction by client in case required. The treated water can be collected either from the chlorination chamber in Scheme I or from Activated Carbon Filter in Scheme II.

Sludge :

The sludge from the Clarifier to be removed from the bottom of the Clarifier once in a day by client and transferred to sludge drying bed either by gravity or through pump depending on site condition. Before starting the Desludging, shut off the aeration for 30 minutes and allow the sludge to concentrate at the bottom. After concentration, the bottom sludge valve can be open to drain the required quantity of sludge. An air scour pipe is provided to dislodge any media choking the drain outlet. Open the air scour valve for short time to remove the clog

1.11 SOLID WASTE MANAGEMENT :

1. The wet waste & dry waste shall be segregated & stored for treatment & disposal to municipal waste disposal authority.
2. The dry waste shall be collected & disposed to the municipal waste disposal authority
3. The wet waste shall be collected & treated in the organic waste composter to convert the kitchen to organic compost to use it as manure for the landscape.
4. The solid waste generated in the project is calculated as below:

SOLID WASTE MANAGEMENT - OPERATIONAL PHASE (BLDG - 01)

Sr. No.	Description	Occupancy (Residential)	Criteria		Solid Waste Generation		Total (Kg / Day)
			Bio-Degradable	Non - Biodegradable	Bio - Degradable (Kg / Day)	Non - Biodegradable (Kg / Day)	
01	Residential Unit	906	0.300	0.200	272	182	454
02	Commercial Unit	136	0.120	0.180	17	25	42
	Total				**289**	**207**	**496**

Sr. No.	Type of Waste	Quantity (Kg / day)	Management
01	Biodegradable Waste	289	Biodegradable waste will be processed in OWC, and manure so obtained will be used for landscaping.
02	Non-biodegradable Waste	207	Will be handed over to recyclers.
03	Garden Waste	-	Will be mixed with biodegradable waste and composted in OWC.

Sr. No.	Description	Area (Sq. Mtr.)
01	For Dry Waste Storage	03
02	For Machine	03
03	For Curing	15
04	For Bio-medical Waste	03
05	For E Waste	03
	Total Area Required	**27**
	Total Area Provided	**30**

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1.12 Site Photos

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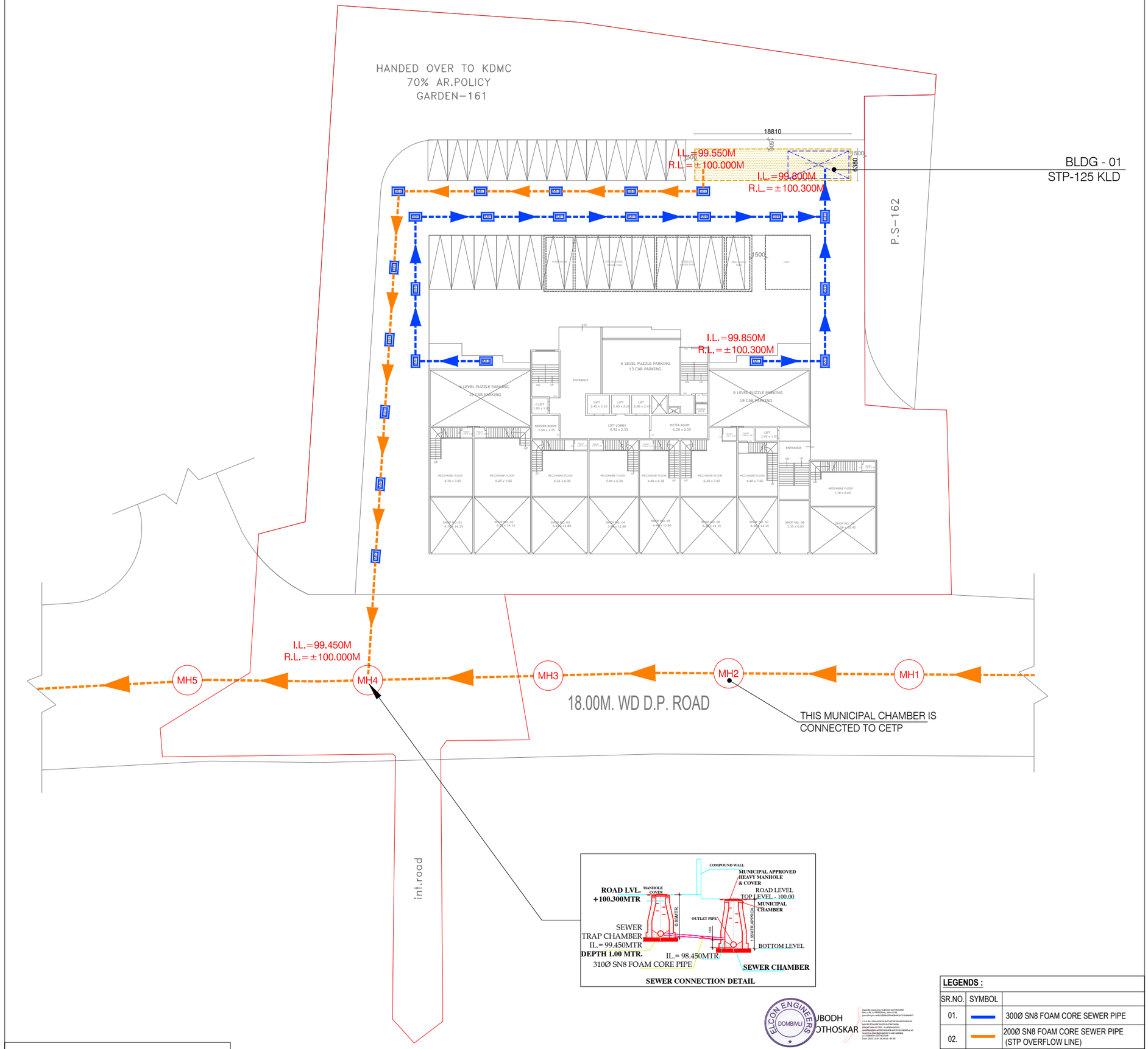
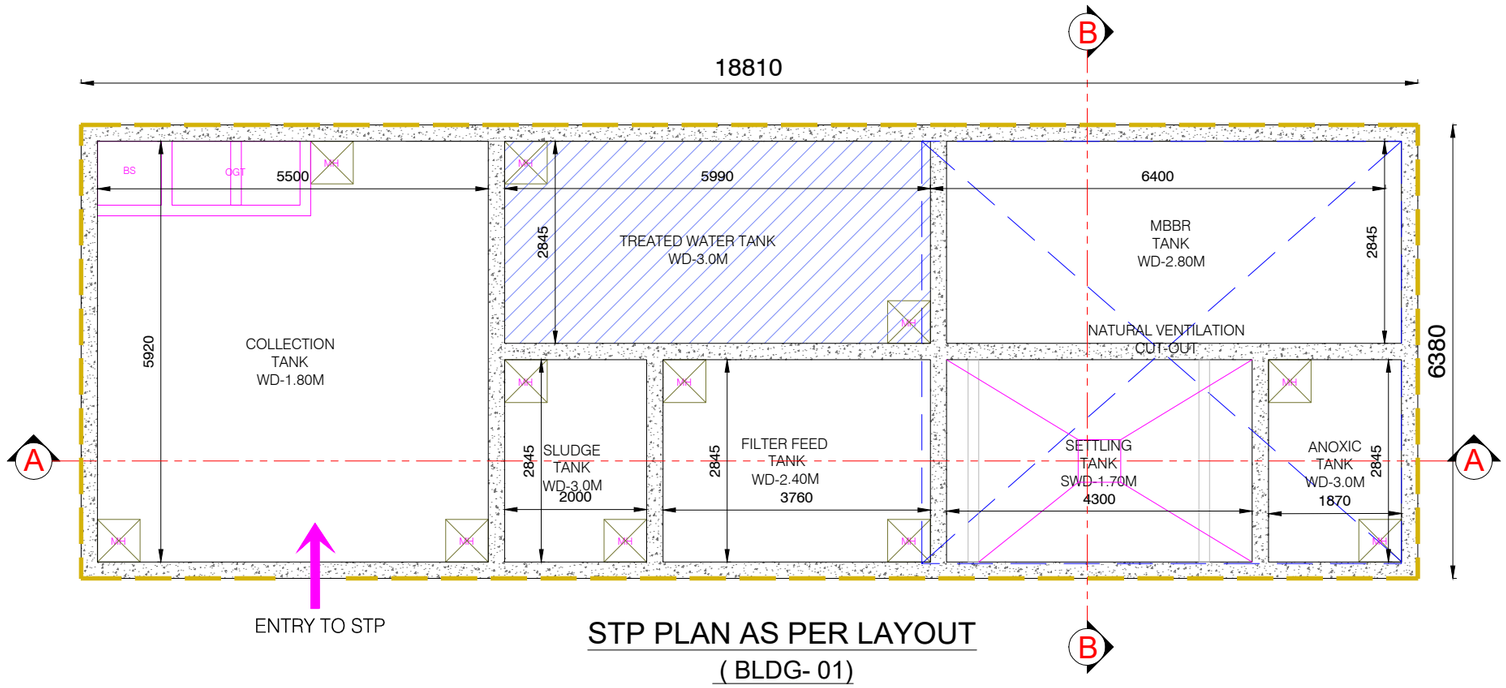


LEGENDS :

SR.NO.	SYMBOL	
01.		300Ø SN8 FOAM CORE SEWER PIPE
02.		200Ø SN8 FOAM CORE SEWER PIPE (STP OVERFLOW LINE)
03.		900X450 RECTANGLE MANHOLE

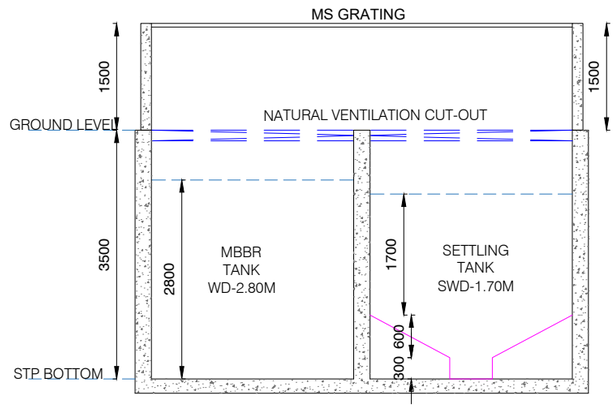
SEWER LINE LAYOUT PLAN

STP DETAILS

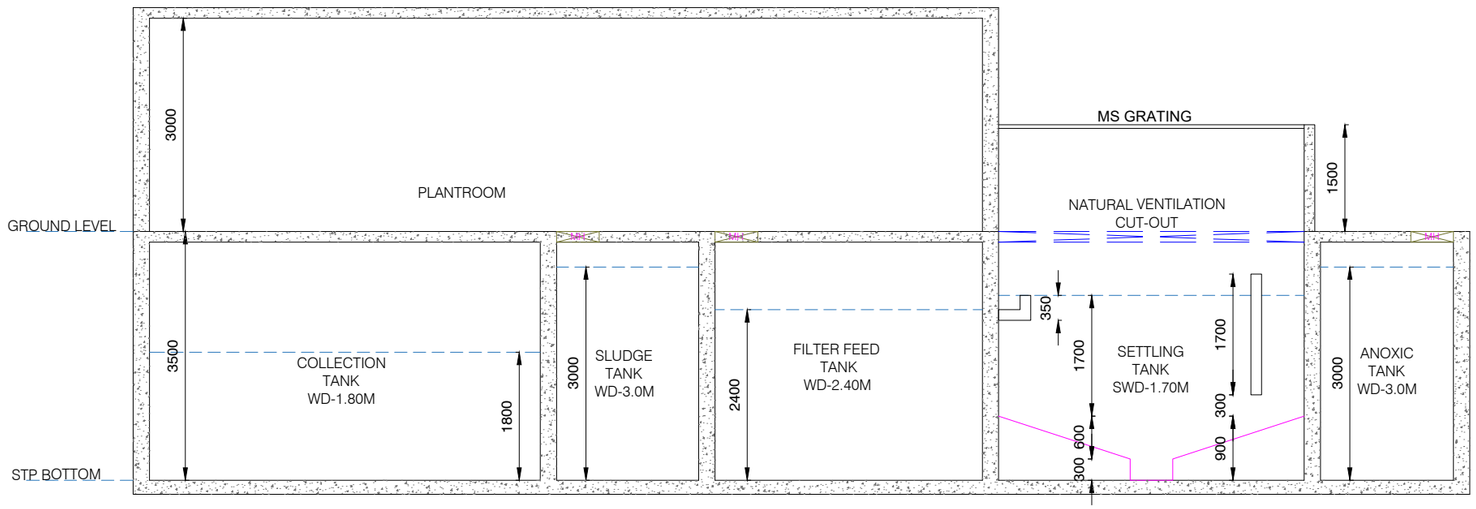


STP PLAN AS PER LAYOUT

(BLDG- 01)



SECTION B-B



SECTION A-A

125 KLD STP @24 HOURS OPERATING PER HOUR 5.21 CU.M/hr									
DESCRIPTION	SIZE			FREE BOARD	RETENTION TIME IN Hrs.	VOLUMN	MOC	POSITION	AREA
	W	L	H						
BAR SCREEN CHAMBER	1.00	1.00	1.00	0.20	0.192	1.00	RCC	UG	1.00
COLLECTION TANK	5.50	5.92	3.50	1.55	21.88	113.96	RCC	UG	32.56
ANOXIC TANK	1.87	2.85	3.50	0.35	3.58	18.65	RCC	AG	5.32
MMBR TANK	6.40	2.85	3.50	0.55	12.25	63.84	RCC	AG	18.20
SETTLER-1	4.30	2.85	3.50	0.75	8.23	42.89	RCC	AG	12.23
SLUDGE TANK	2.00	2.85	3.50	0.35	3.83	19.95	RCC	AG	5.70
FILTER FEED TANK	3.76	2.85	3.50	0.95	7.20	37.51	RCC	UG	10.72
TREATED WATER TANK	5.99	2.85	3.50	0.50	11.47	59.75	RCC	UG	17.07

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SUBODH GOTHOSKAR

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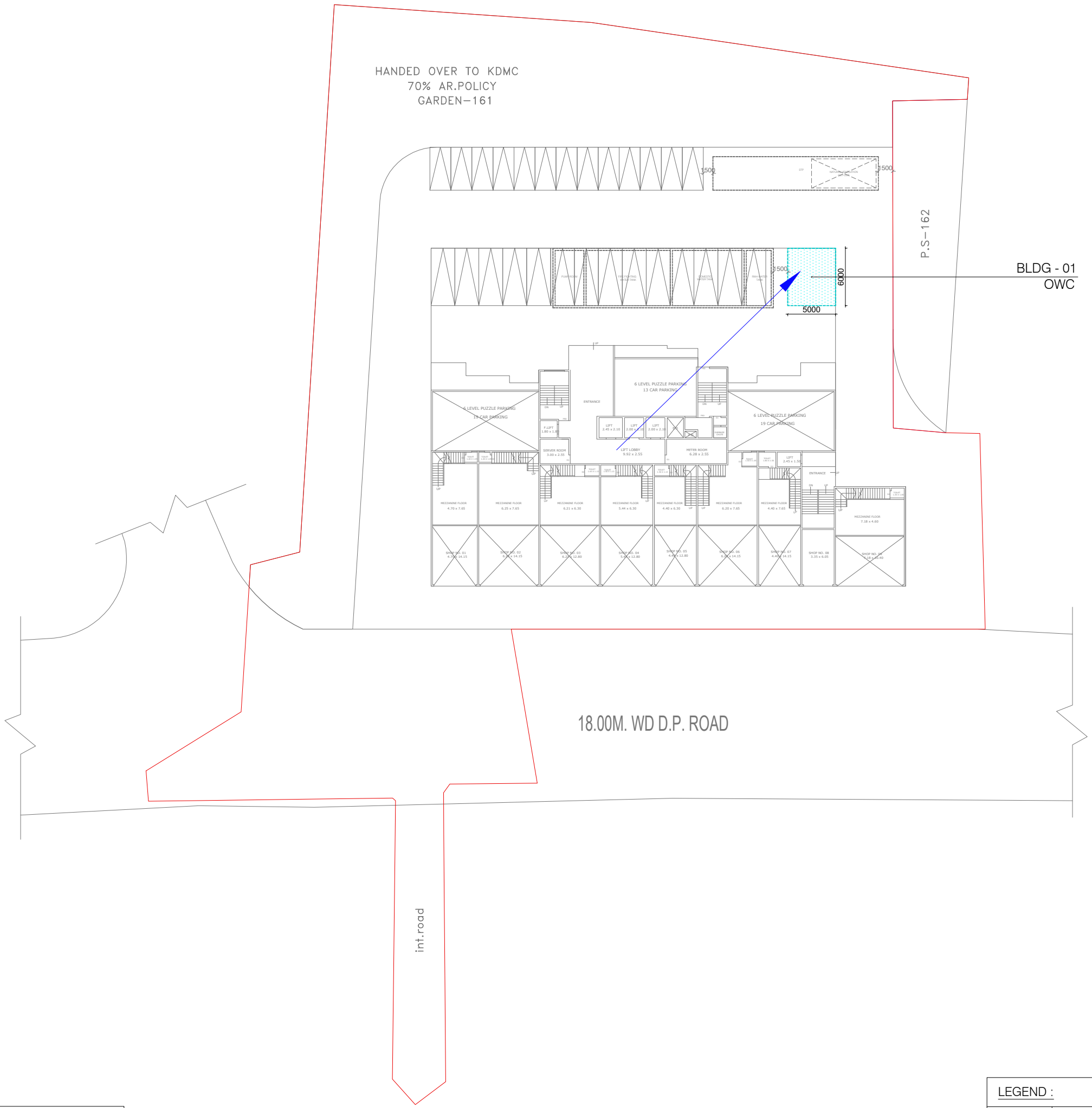

O.W.C LAYOUT PLAN

LEGEND :

	O.W.C.

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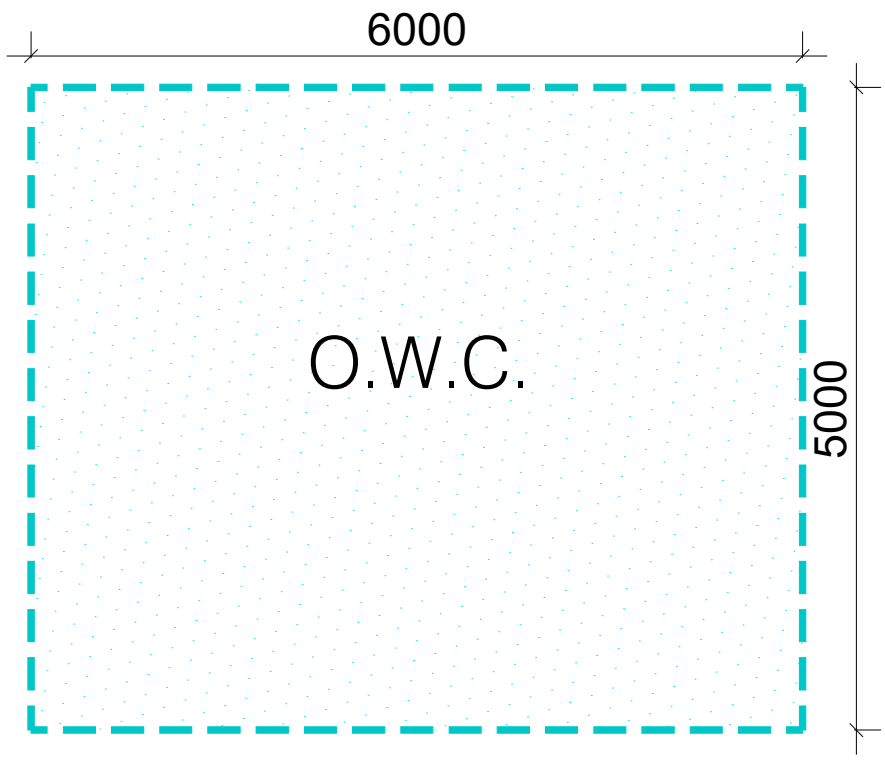
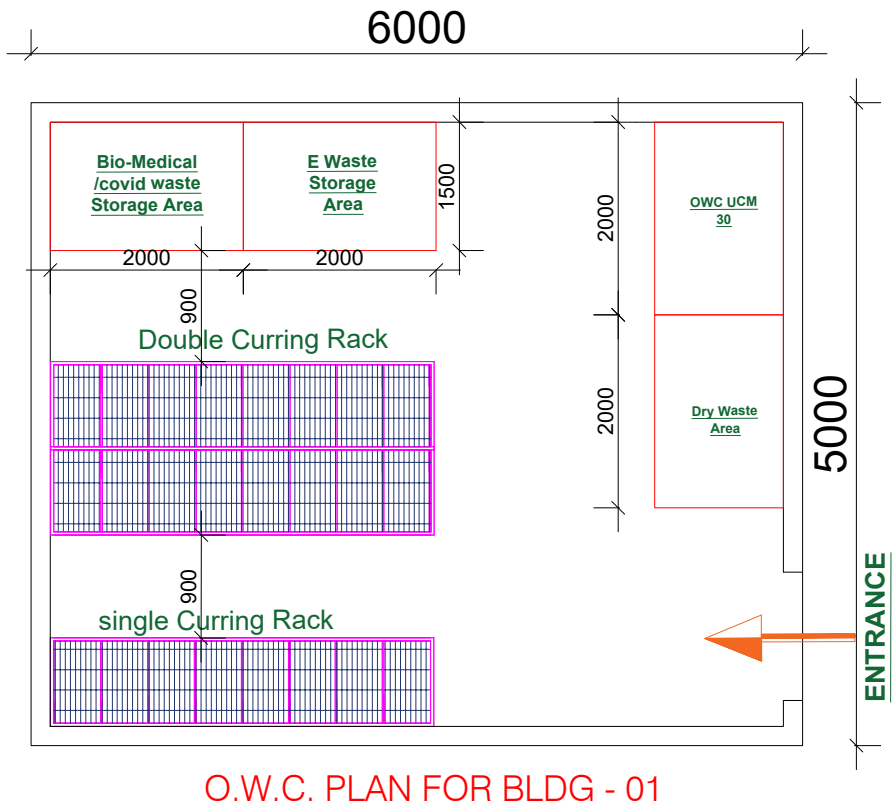
OTHOSKAR

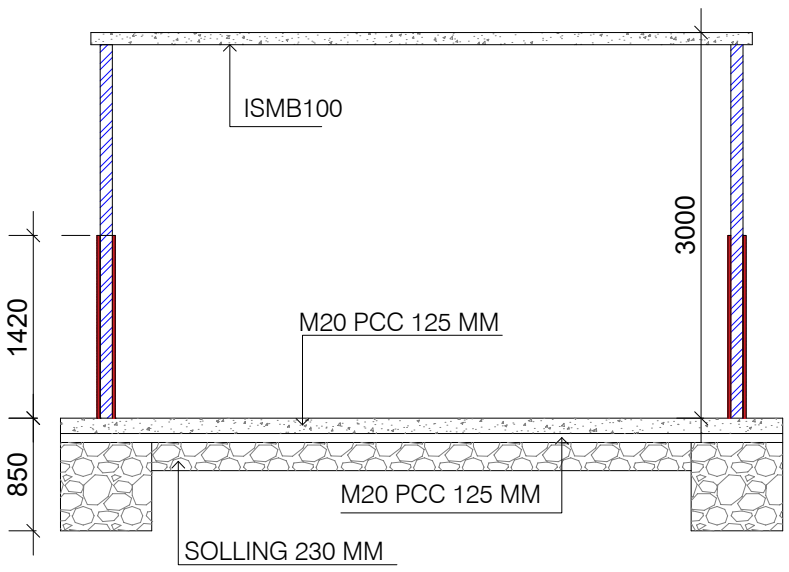
O.W.C.DETAILS - BLDG - 01



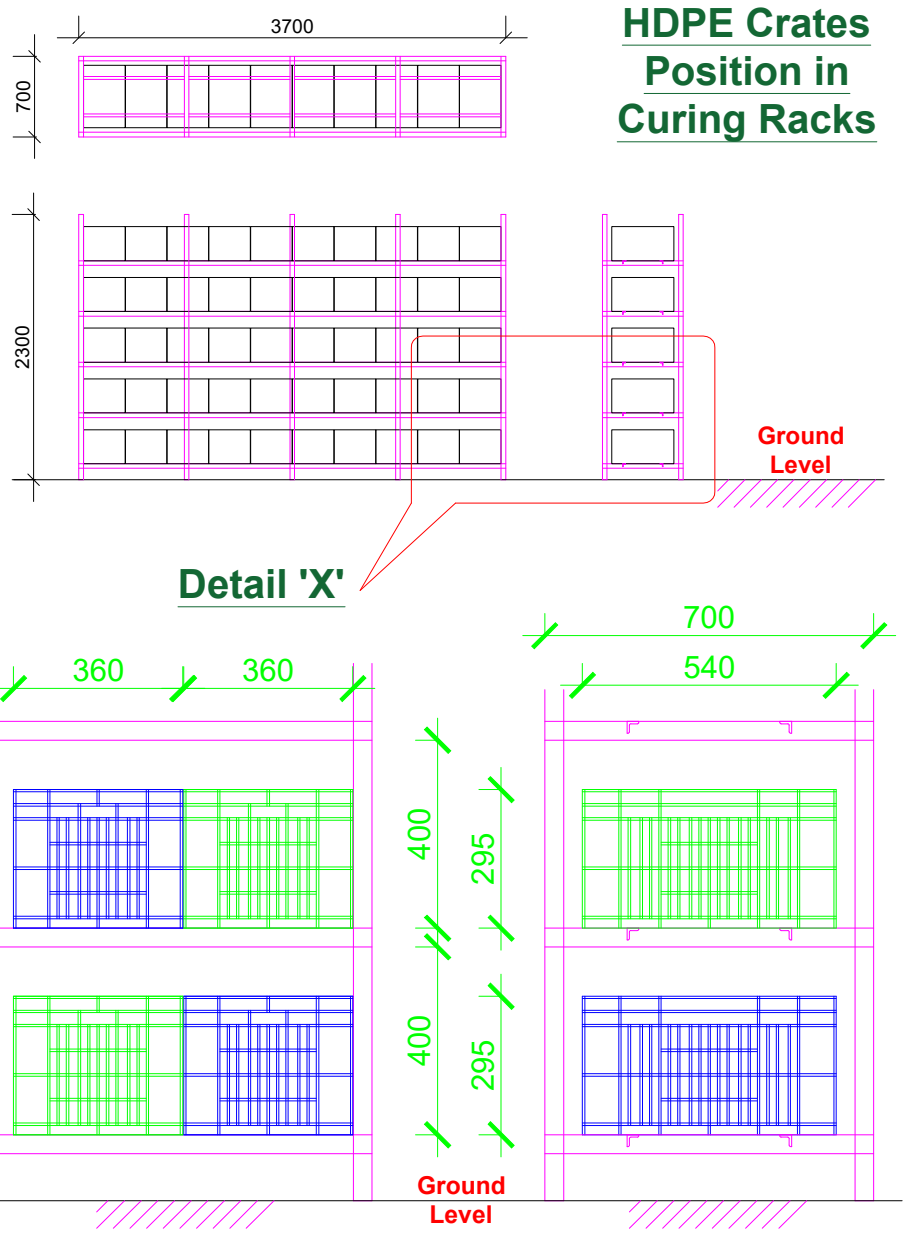
O.W.C. PLAN FOR BLDG - 01
AS PER LAYOUT



O.W.C. PLAN FOR BLDG - 01
AS PER LAYOUT



SECTION AT A-A



Enlarged View of Detail 'X' (Crates Position in Curing Racks)

SR.No.	Building No.	REQUIRED AREA (IN SQM.)	PROVIDED AREA (IN SQM.)
1	FOR DRY WASTE STORAGE	3.00 SQ.M.	3.00 SQ.M.
2	FOR OWC MACHINE	3.00 SQ.M.	3.00 SQ.M.
3	FOR CURRING PROCESS (RACKS)	10.00 SQ.M.	18.00 SQ.M.
4	FOR BIO-MEDICAL & HAZARDOUS WASTE	3.00 SQ.M.	3.00 SQ.M.
5	FOR E-WASTE	3.00 SQ.M.	3.00 SQ.M.
	TOTAL AREA REQUIRED	22.00 SQ.M.	30.00 SQ.M.

OWC Plant Location for BLDG - 01

Building No.	OWC Machine Model	Curing Racks Qty.		Bio-degradable Waste Qty.	Non Bio-degradable Waste Qty.
		Single	Double		
BLDG - 01	UCM 30	01	01	289 Kgs/Day	207 Kgs/Day

Notes:

1. Three Phase Power Supply up to 5 kW load is required in the Shed. with Proper Earthing connection.
2. Cross Ventilation is must in the shade, if not possible then Proper Exhaust Duct should be provided.
3. Water Tap provision to be available in the shed.
4. Proper Shed need to be build it proposed in open area.

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BODH THOSKAR

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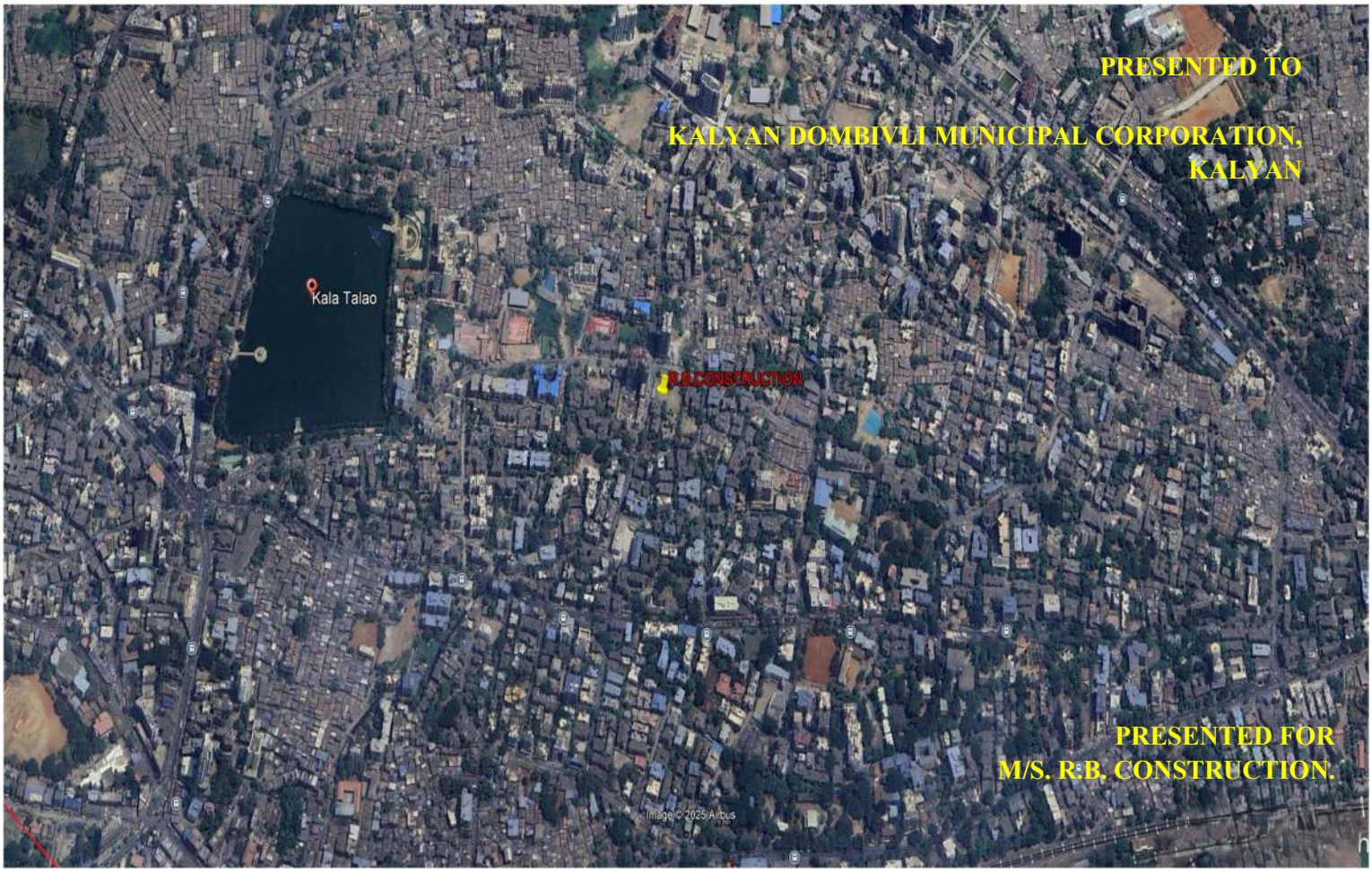
STORM WATER MANAGMENT PLAN FOR PROPOSED RESI. & COMM. BUILDING ON PLOT BEARING S.NO. S. No. 105/8/2, 97/1/9 & 98/2/5 OF MOUJE - CHIKANGHAR, TAL. -KALYAN, DIST. – THANE.



DECEMBER 2025

PREPARED BY

ELCON ENGINEERS

SUBODH GOTHOSKAR

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Date: 2025.12.01 18:19:45 +05'30'

ELCON ENGINEERS DOMBIVLI

Date : 01.12.2025

Ref: Elcon/M/s. R.B. Construction./SWD/2425-149/**02**

To,
M/s. R.B. Construction.
Kalyan.

Sub: Design of stormwater drain for proposed Resi. & Comm. building on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane.

Dear Sir,

We are submitting design of storm water drain for proposed Resi. & Comm. building on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane. The storm water drain has been designed for the rainfall intensity of 150 mm/h. KDMC benchmark levels has been followed for designing storm water drains. The storm water from the plot is proposed to be disposed into an existing roadside storm water drain passing front of the plot.

Thanking You,
For Elcon Engineers

SUBODH GOTHOSKAR

ELCON ENGINEERS DOMBIVLI

Authorised Signatory

Sr. No.	Content	Page No.
1.	Summary	1
2.	Background	2
3.	Goals of Storm Water Management Plan	2
4.	Design & Performance Standards	2
5.	Recommendations	3
6.	Annexure	3
	Annexure I - Design Calculations	
	Annexure II - Drawing Plates	

1. Summary

This report has been prepared by Elcon Engineers on behalf of M/s. R.B. Construction, with a guideline set out by Kalyan Dombivli Municipal Corporation. The following table is an overview of the storm water management plan for the proposed development of the site.

Sr. No.	Content	Response
1.	Site Location	Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane.
2.	Proposed Development	Construction of the Residential Cum Commercial Building plan accommodation policy
3.	Historical Flooding	None
4.	Flood Risk	No
5.	Design Rainfall Intensity	150mm/h
6.	Sustainable Urban Drainage System (SUDS) Measures	1. Roof Top Rainwater is Proposed to be collected in a rain water collection tank to reduce the peak flow & Discharge of storm water in a municipal nalla.

2. Background

The proposed Resi. & Comm. building on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane. The site lies within the Kalyan Dombivli Municipal Corporation jurisdiction. Consequently, the existing land use pattern of the proposed plot and the nearby area is undergoing redevelopment and built-out that is resulting in increased storm water runoff. This increased storm water volume eventually discharges into the main drains. The storm water management plan documents the strategy to redress storm water issues related to development. This plan is being prepared to bring the proposed site into compliance with Kalyan Dombivli Municipal Corporation storm water regulations. The storm water management plan addresses disposal of storm water and groundwater recharge by incorporating storm water design.

Figure 1 locates the plot on Google map.

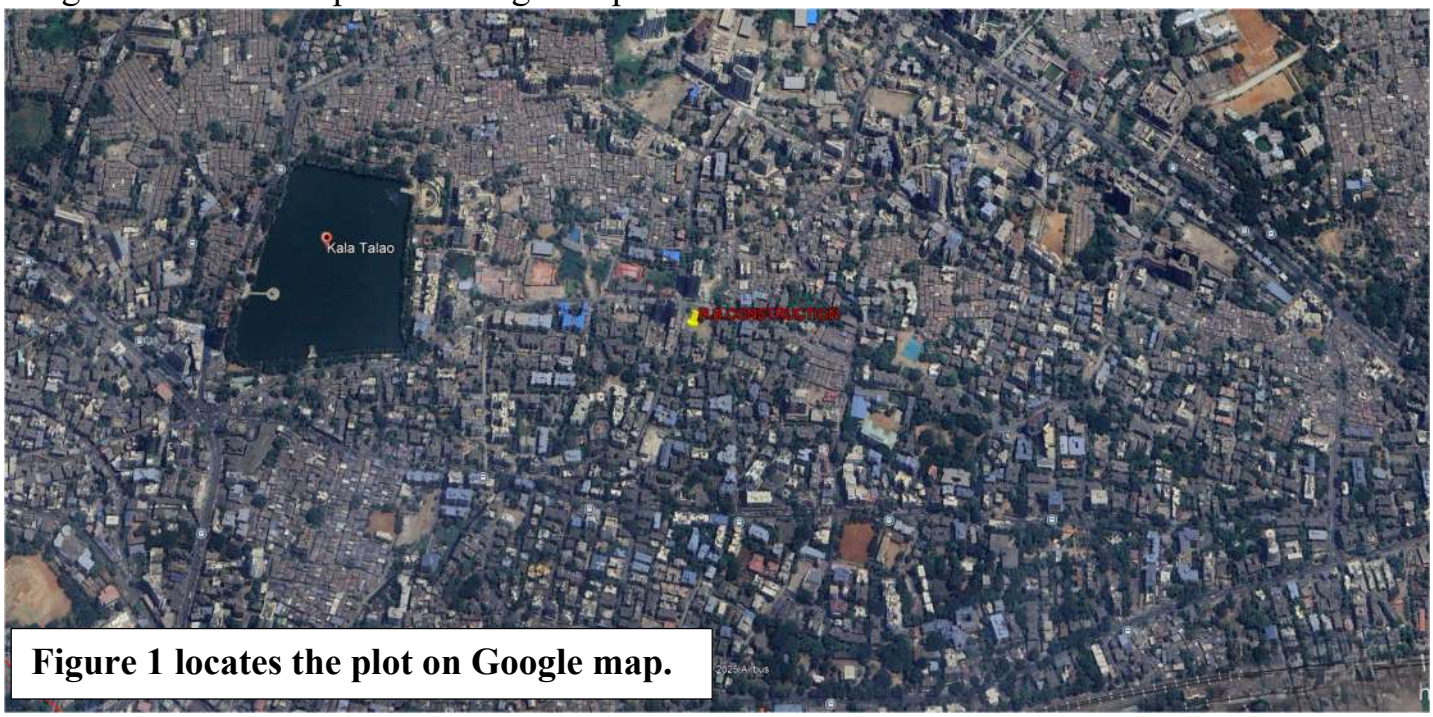

Figure 1 locates the plot on Google map.

3. Goals of storm water management plan

Through this storm water management plan, the client seeks to establish a guiding document with the intended goals of:

1. Effective disposal of storm water.

4. Design and performance standards.

Through this storm water management plan, the client seeks to establish a guiding document with the intended goals of:

1. All drains have been designed to carry runoff generated from the rainfall intensity of 150 mm/h with coefficient of runoff of 1.00 for impervious and previous areas.
2. Self –cleansing velocity (velocity>0.80) has been maintained to prevent the deposition of silt in drains.
3. The storm water from the plot is proposed to be disposed into an existing roadside storm water drain passing front of the plot.

5. Recommendations

The storm water from the plot is proposed to be disposed into an existing road site storm water drain passing front of the plot.

Annexure I

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Design of storm water drain for proposed Resi. & Comm. building on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane.

Rainfall Intensity = 150 mm/h

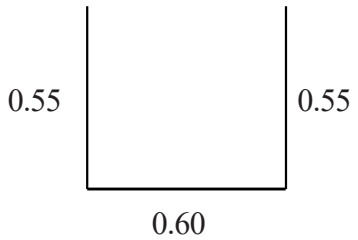
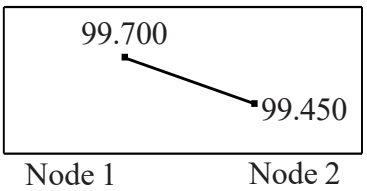
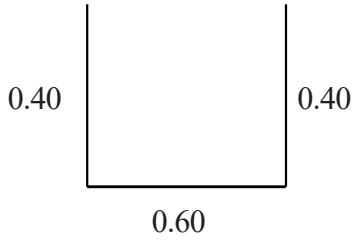
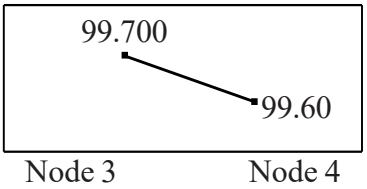
Coefficient of runoff for pervious area and impervious areas = 1.0

Node	Q1 Initial (m^3/sec)	Contributing area (sq. m)		Q2 Impervious (m^3/sec)	Q3 pervious (m^3/sec)	Cumulative Q (m^3/sec)	Width (M)	Depth (M)	Depth including free board (M)	Wetted area A (sq.m)	Wetted Perimeter P (M)	R	Slope, S	N	Velocity (M/S)	Q full (M3/sec)	Remark
		Impervious	Pervious (Garden)														
1 to 2	0.00	500	0.00	0.021	0.00	0.02	0.60	0.55	0.70	0.42	2.00	0.21	1:300	0.013	1.57	0.66	OK
3 to 4	0.00	600	0.00	0.025	0.00	0.03	0.60	0.40	0.55	0.33	1.70	0.19	1:300	0.013	1.49	0.49	OK

Elcon Engineers

Design of storm water drain for proposed Resi. & Comm. building on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane.

Storm Water Drain Design Summary

Node 1 To Node 2		
Node 3 To Node 4		

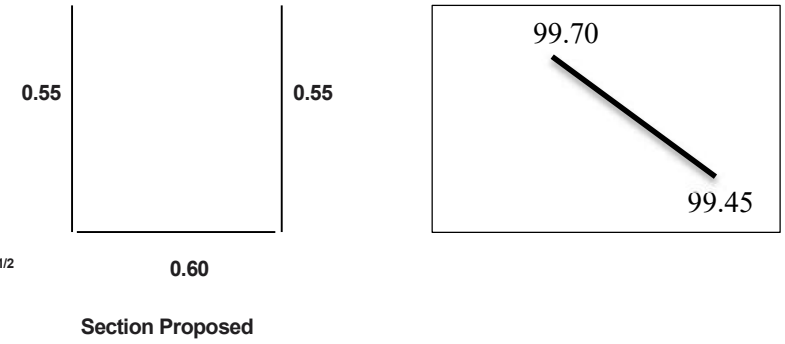
Design of storm water drain for proposed Resi. & Comm. building on Plot Bearing S. No. 105/8/2, 97/1/9 & 98/2/5 of Mouje - Chikanghar, Tal- Kalyan, Dist.-Thane.

Node 1 to Node 2

Initial flow (Q1)	= 0.00
Contributing area , A (impervious)	= 0.00050 sq.mm
Coefficient of runoff, C	= 1
Intensity of rainfall, 1	= 150 mm/hr
Q2	= CIA/3.6 m^3/ sec
	= 0.02 m^3 / sec
Contributing area, B (Pervious)	= 0.00050 sq.mm
Coefficient of runoff, C	= 1
Intensity of rainfall, 1	=150mm /hr
Q3	= CIA/3.6m^3 / sec

Drain Provided

Width W	= 0.60 m
Depth, D	= 0.55 m
Slope, S	= 1/300
	= 0.0033
N	= 0.013
Velocity, V	= 1/ $nR^{2/3}$ $S^{1/2}$
	= 1.57 m/s
Q provided	= 0.60 x 0.55 x 1.57
	= 0.66 m^3 / sec > 0.02 (Q1+Q2+Q3) m^3 / sec



Section Proposed

Section provided is adequate

Section Proposed

Width, W	= 0.60 m
Depth D	= 0.55 m

Node 3 to Node 4

Initial flow (Q1) = 0.00

Contributing area , A (impervious) = 0.00060 sq.mm

Coefficient of runoff, C = 1

Intensity of rainfall, 1 = 150 mm/hr

Q2 = CIA/3.6 m^3/ sec

= 0.03 m^3 / sec

Contributing area, B (Pervious) = 0.00060 sq.mm

Coefficient of runoff, C = 1

Intensity of rainfall, 1 =150mm /hr

Q3 = CIA/3.6m^3 / sec

Drain Provided

Width W = 0.60 m

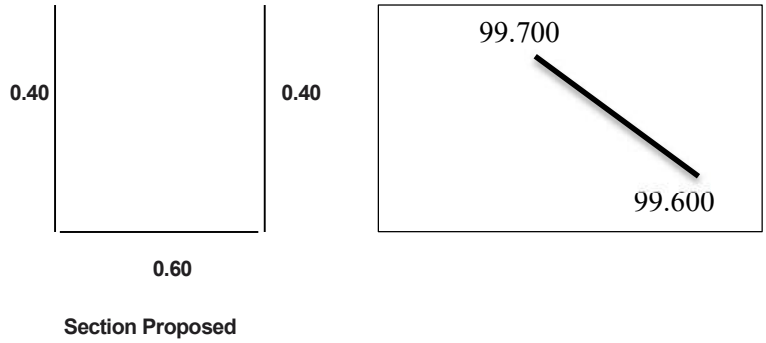
Depth, D = 0.40 m

Slope, S = 1/300

= 0.0033

N = 0.013

Velocity, V = 1/ $nR^{2/3}$ $S^{1/2}$



Section Proposed

= 1.49 m/s

Q provided = 0.60 x 0.40 x 1.49

= 0.49 m^3 / sec > 0.03 (Q1+Q2+Q3) m^3 / sec

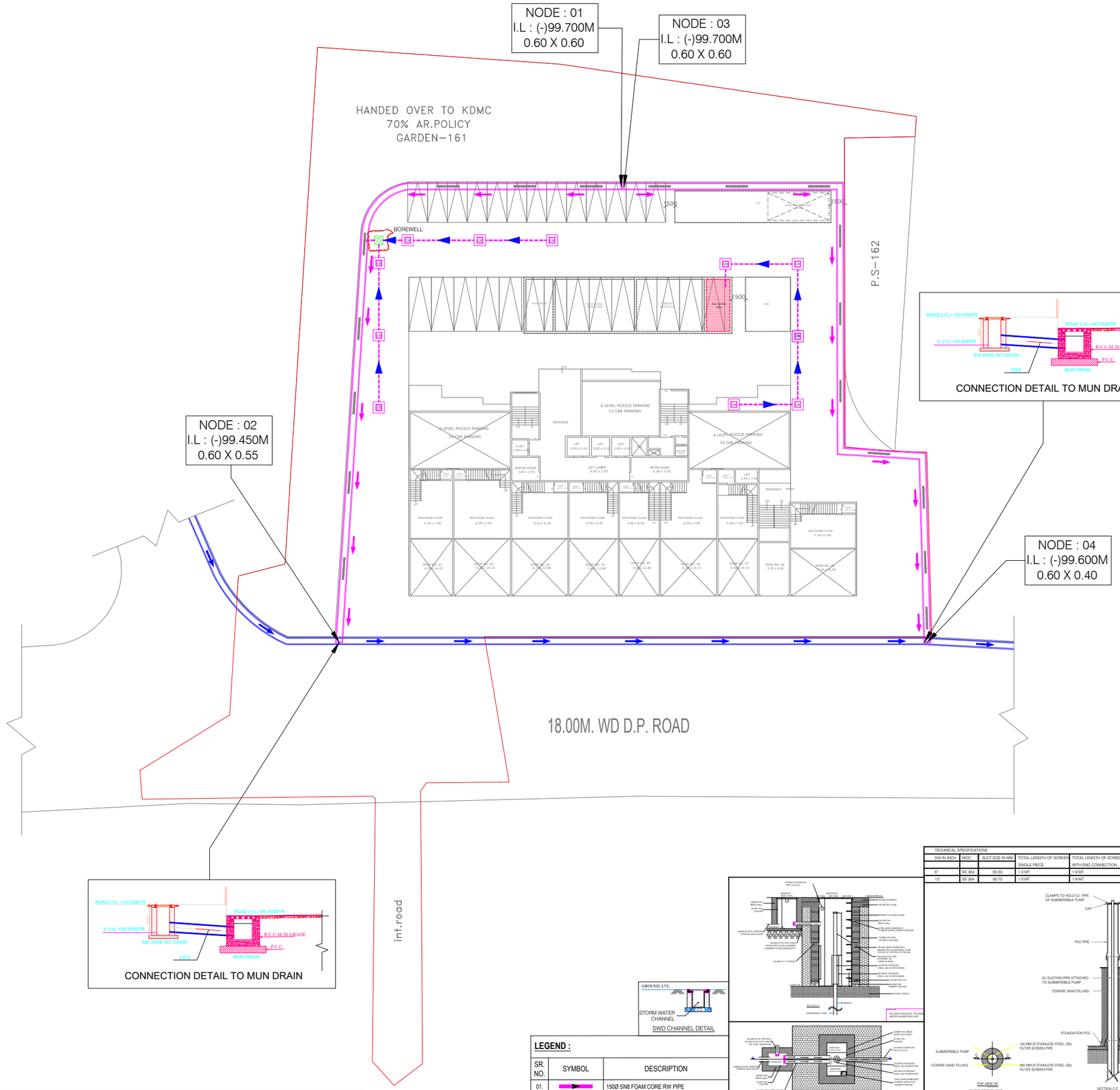
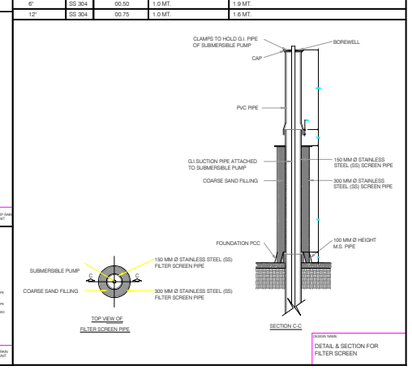
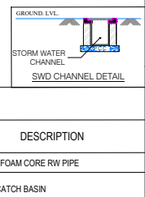
Section provided is adequate

Section Proposed

Width, W = 0.60 m

Depth D = 0.40 m

Annexure II



TECHNICAL SPECIFICATIONS				
DIA IN INCH	MOC	SLOT SIZE IN MM	TOTAL LENGTH OF SCREEN SINGLE PIECE	TOTAL LENGTH OF SCREEN WITH END CONNECTION
6"	SS 304	00.50	1.0 MT.	1.9 MT.
12"	SS 304	00.75	1.0 MT.	1.6 MT.